Lecture 14

Faraday's Law
Induced EMF

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The Force From a **Static** EM Field is Conservative

- Definition of Voltage

$$V_B - V_A = -\int_A^B \mathbf{E} \bullet d\mathbf{l}$$

Voltage drop from B to A is equal to the work you need to do to move a unit charge from A to B against the electric field $\mathbf{E}$.

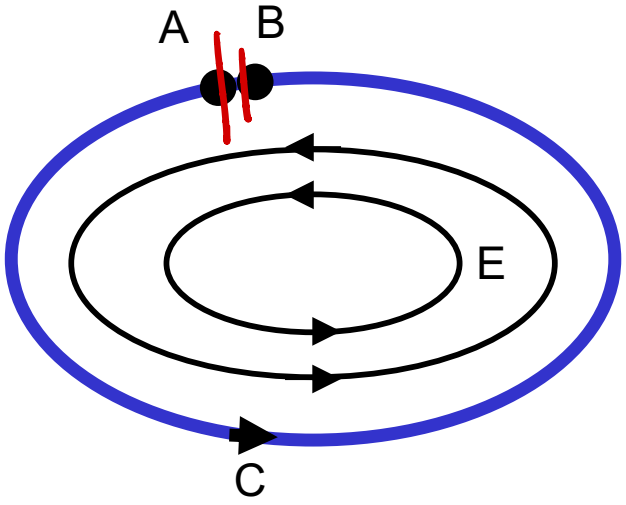
The following are equivalent:

$\vec{\boldsymbol{E}}$ is conservative $\vec{\boldsymbol{E}}$ is curl-free

$$\oint_C \vec{\boldsymbol{E}} \cdot d\vec{\boldsymbol{l}} = 0 \qquad \int_A^B \vec{\boldsymbol{E}} \cdot d\vec{l} \text{ is path independent}$$

$$\vec{E} = -\nabla V$$

Non-conservative Fields: Electromotive Force (emf)



EMF, Voltage / electric potential. [V].

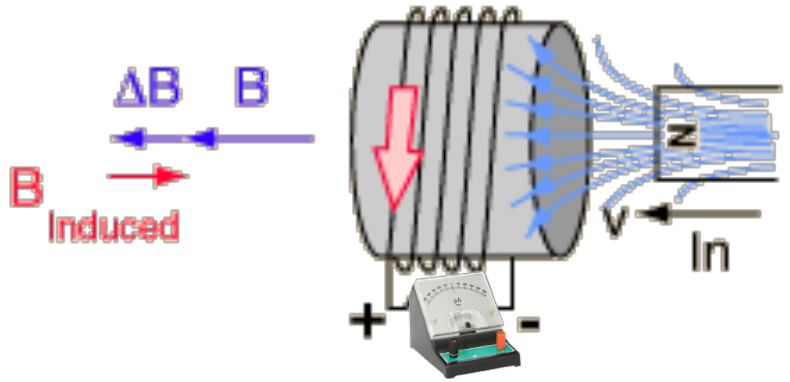
$$\mathcal{E} = \oint_C \vec{\boldsymbol{E}} \cdot d\vec{\boldsymbol{l}} = -\frac{d\Psi}{dt}$$

= Line integral of E around the closed path C (counter-clockwise).

EMF can be non-zero if EM field varies in time.
EMF is a difference in potential that can give rise to an electric current. Think of it as a battery between A and B.

Faraday's Law

When the magnetic flux enclosed by a loop of wire CHANGES WITH TIME, a current is produced in the loop



The EMF in the loop is the NEGATIVE of the rate of change of the magnetic flux enclosed in the loop

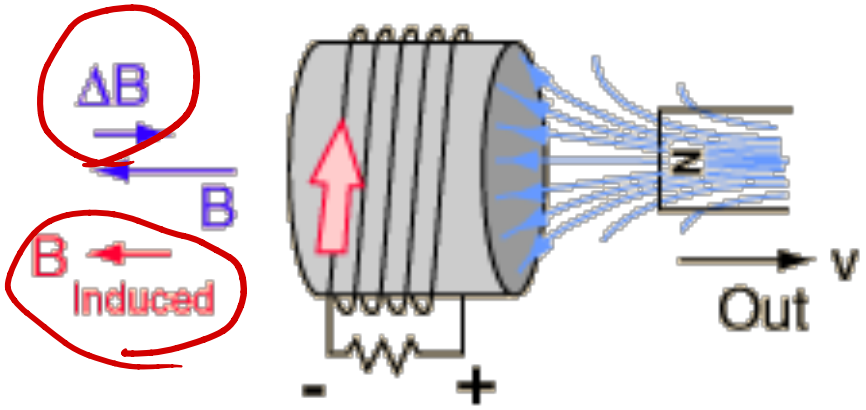
$$\oint_C \vec{\boldsymbol{E}} \cdot d\vec{\boldsymbol{l}} = -\frac{d}{dt}\int_S \vec{\boldsymbol{B}} \cdot d\vec{\boldsymbol{S}}$$

Magnetic flux $\Psi \equiv \int_S \vec{\boldsymbol{B}} \cdot d\vec{\boldsymbol{S}}$ [Weber (Wb)]

Unit of $\vec{\boldsymbol{B}}$ is $\mathrm{Wb/m^2}$ or Tesla (T)

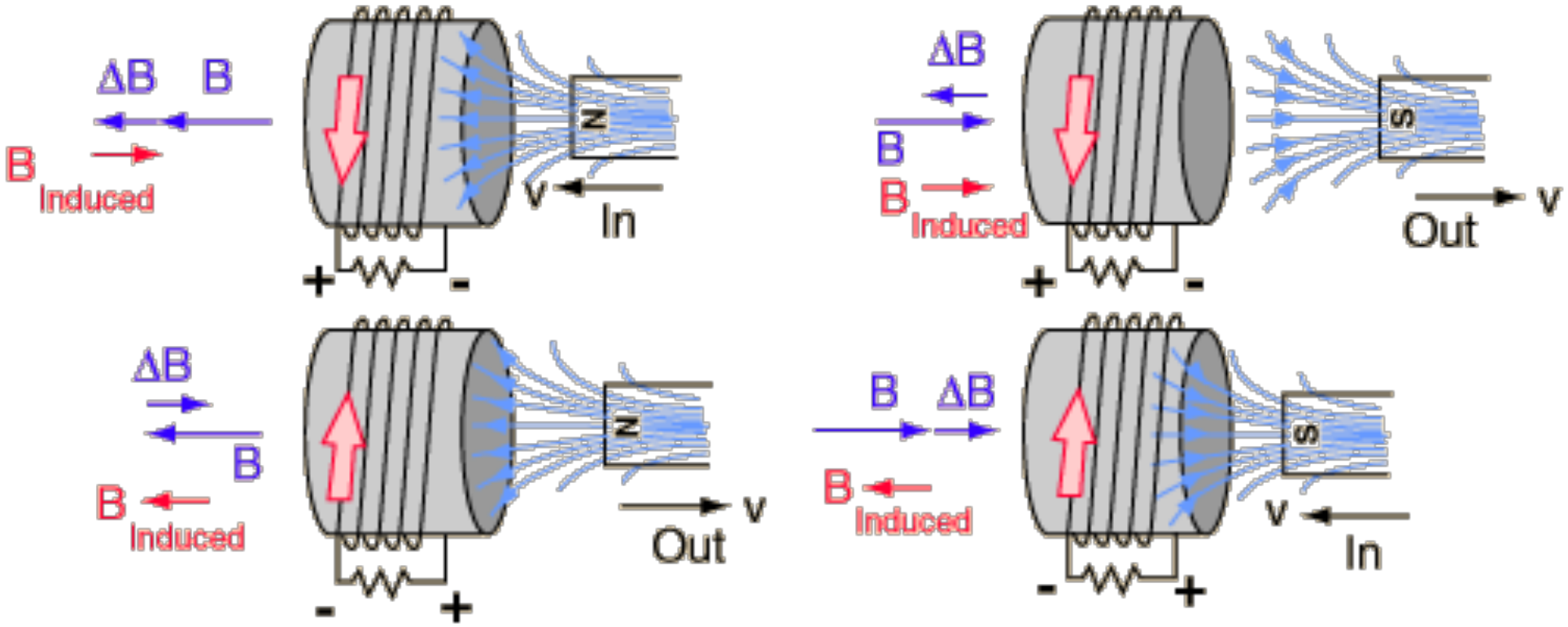
Lenz's Law

- The direction of the induced EMF always OPPOSES the CHANGE in magnetic flux that produces it.
- It opposes the **CHANGE** in flux, not the flux itself!



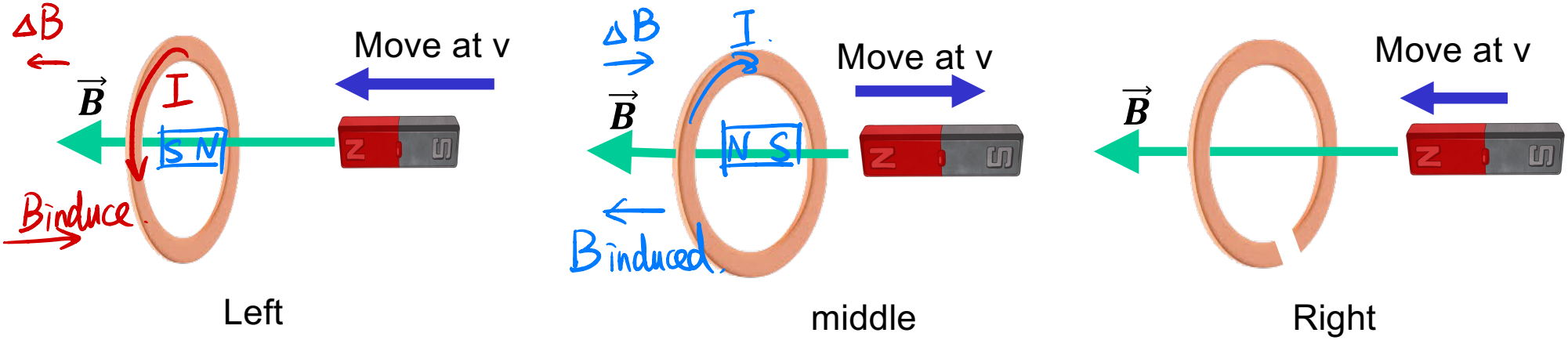
  - Explains why it is "-d/dt"

Experiments for Faraday's Law



Lenz's Law Demonstration $\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$

- The direction of the induced EMF always OPPOSES the CHANGE in magnetic flux that produces it.

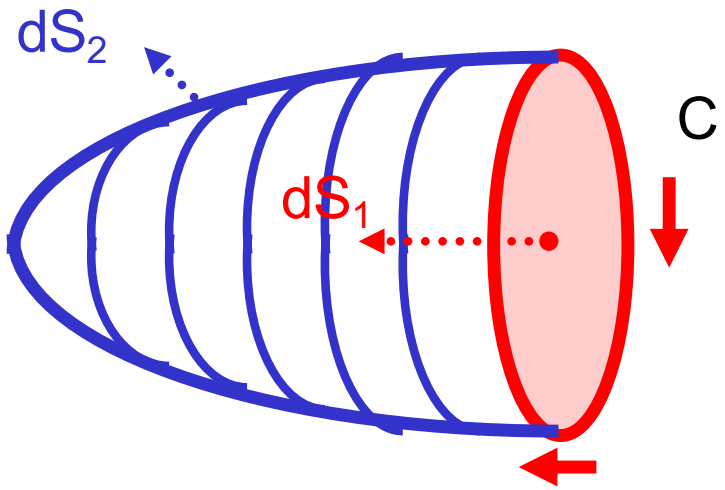


Left middle Right

Discussion:

1. If the magnet moves into a copper ring (left figure), will the copper ring move left or right?
2. When the magnet leaving the copper ring (middle figure), will the copper ring move left or right?
3. If the copper ring has a cut (right figure), will the it move left or right?
4. Does it matter if the magnet has N or S to the left?

Faraday's Law Rules



$$\iint_S \nabla\times\vec{E}\cdot d\vec{S}$$

$$\oint_C \vec{\boldsymbol{E}}\cdot d\vec{\boldsymbol{l}} = -\frac{d}{dt}\int_S \vec{\boldsymbol{B}}\cdot d\vec{\boldsymbol{S}}$$

$$\nabla\times\vec{\boldsymbol{E}} = -\frac{d\vec{\boldsymbol{B}}}{dt}$$

- Right Hand Rule
  - Right hand curls around C so thumb points in direction of $d\vec{\boldsymbol{S}}$

Induced emf

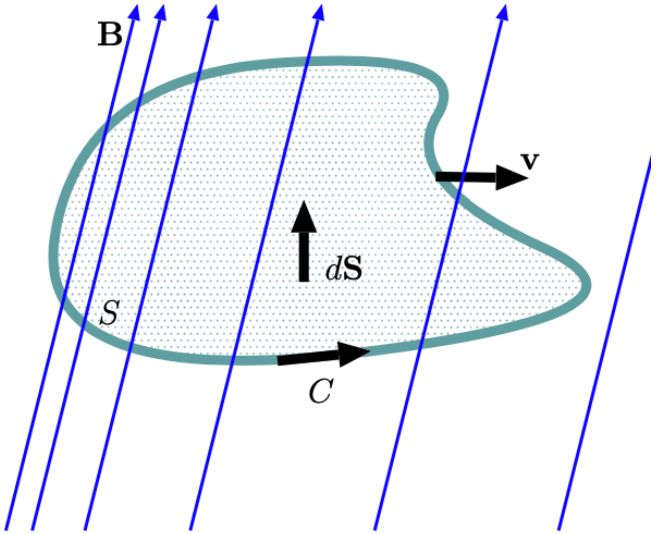
If C is stationary

$$\mathcal{E} = \oint_C \vec{\boldsymbol{E}} \cdot d\vec{\boldsymbol{l}} = -\frac{d\Psi}{dt} = -\frac{d}{dt}\int_S \vec{\boldsymbol{B}} \cdot d\vec{\boldsymbol{S}}$$

EMF [V]

If C is in motion

$$\mathcal{E} = -\frac{d\Psi}{dt} = \oint_C \left(\vec{\boldsymbol{E}} + \vec{\boldsymbol{v}} \times \vec{\boldsymbol{B}}\right) \cdot d\vec{\boldsymbol{l}}$$



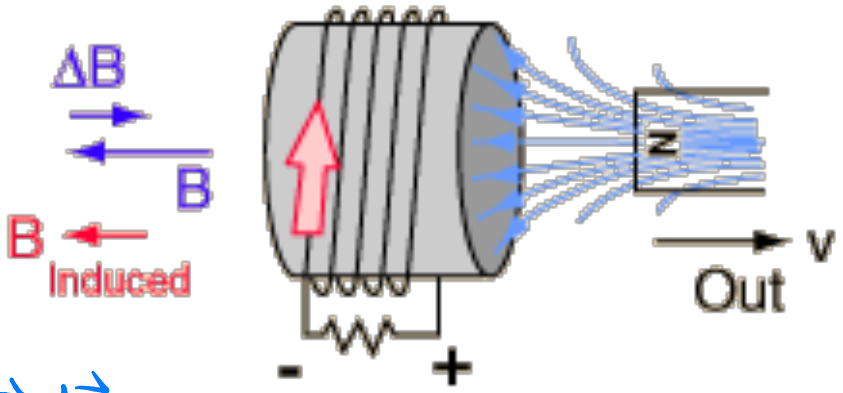
if C is a physical conducting path with a total resistance R then the emf will drive a steady-state or quasi-static current

$$I = \frac{\mathcal{E}}{R}$$

Faraday's Law Rules

- EMF increases in proportion to the number of turns of wire
  - If the loop, C, contains N turns of wire, the EMF is multiplied by N

$$emf = \mathcal{E} = -N\frac{d\Psi}{dt}$$

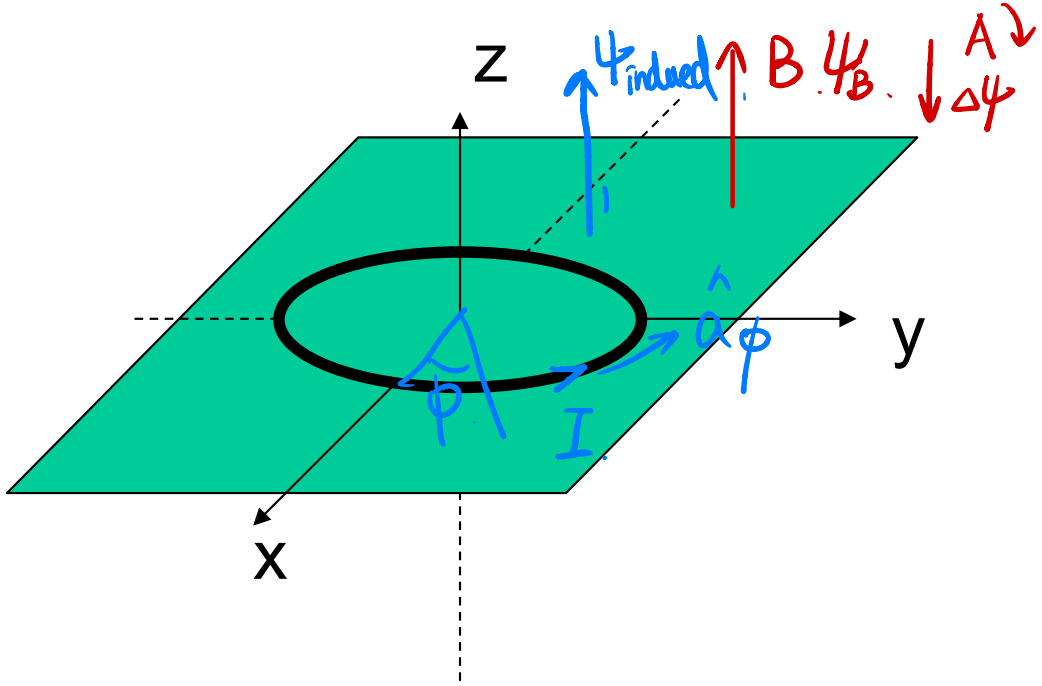


Ψ = single wire $\iint_S \vec{B} \cdot d\vec{s}$

Challenge Question

- A loop, radius R, centered at the origin, sits in the xy plane in the presence of a uniform field $\mathbf{B}=B_0\mathbf{a}_z$ ($B_0>0$). If the loop radius begins to decrease, which direction is the induced current?

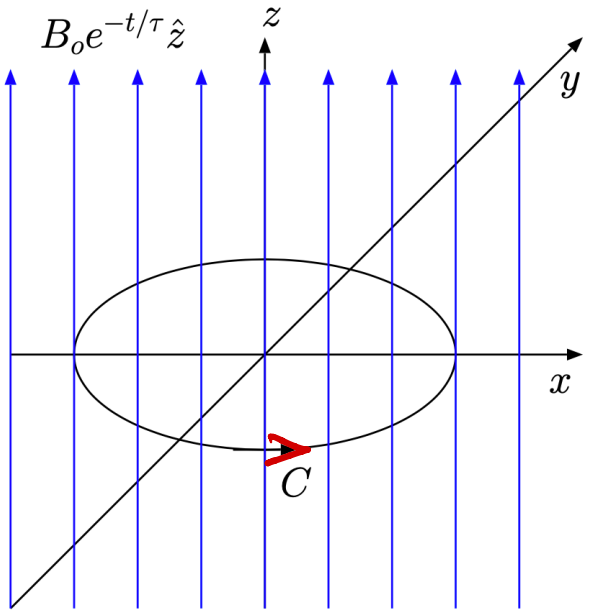
(a) $\mathbf{a}_\phi$
(b) $-\mathbf{a}_\phi$
(c) $\mathbf{a}_z$
(d) cannot be determined



Example: Magnetic Flux and EMF

Given $\vec{\boldsymbol{B}} = B_0 e^{-t/\tau}\hat{z}$

What is the emf ℰ taken over a stationary circular loop C of radius r on z = 0 plane in counter-clockwise direction (looking down on z = 0 plane)? What is current I if the loop resistance is R?



Steps:

1. Write down Faraday's law
2. Write down expression for dS. DIRECTION!!
3. Perform dot product B*dS
4. Solve double integral over limits of the loop
5. Take time derivative of result. Put in "-" sign!

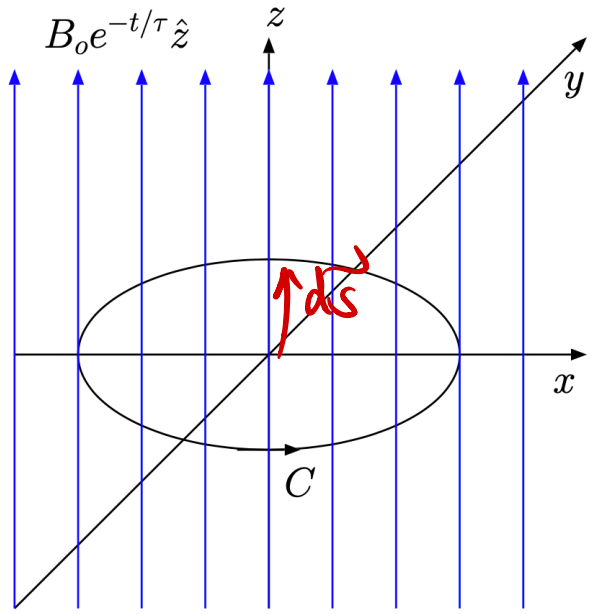
$$EMF = \oint_C \vec{E}\cdot d\vec{l} = -\frac{d}{dt}\iint_S \vec{B}\cdot d\vec{s}$$

$\vec{E}$ → $(\vec{E}+\vec{v}\times\vec{B})$ moving C.

Example: Magnetic Flux and EMF

Given $\vec{B} = B_0 e^{-t/\tau}\hat{z}$

What is the emf $\mathcal{E}$ taken over a stationary circular loop C of radius r = 10 m on z = 0 plane in counter-clockwise direction (looking down on z = 0 plane)? What is current I if the loop resistance is R?



$$\mathcal{E} = EMF = \oint_C \vec{E}\cdot d\vec{l} = -\frac{d}{dt}\iint_S \vec{B}\cdot d\vec{s}$$

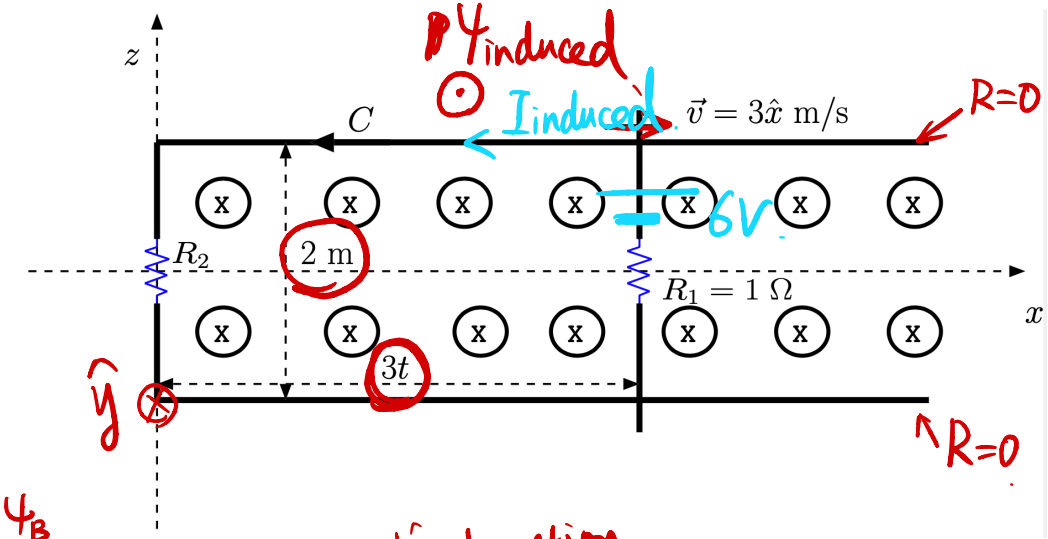
$$\psi_B = \iint_S \vec{B}\cdot d\vec{s} = \iint_S B_0 e^{-t/\tau}\hat{z}\cdot dx\,dy\,\hat{z} = B_0 e^{-t/\tau}\cdot \pi r^2$$

$$\mathcal{E} = -\frac{d\psi_B}{dt} = \frac{B_0 e^{-t/\tau}}{\tau}\cdot \pi r^2 \quad [V]$$

$$I = \frac{\mathcal{E}}{R} \quad [A]$$

Example: Motional EMF

A conducting bar of resistance $R_1 = 1\ \Omega$ is moving in the x-direction with a velocity $\vec{v} = 3\hat{x}$ m/s on a pair of PEC stationary rails. The rails are separated by 2 m and terminated with a load resistance R_2 at $x = 0$. Magnetic field is constant $\vec{B} = 1\hat{y}$ T. Find induced EMF in the loop.



$$\mathcal{E} = \oint_C (\vec{E} + \vec{v} \times \vec{B})\, d\ell = -\frac{d\psi_B}{dt}$$

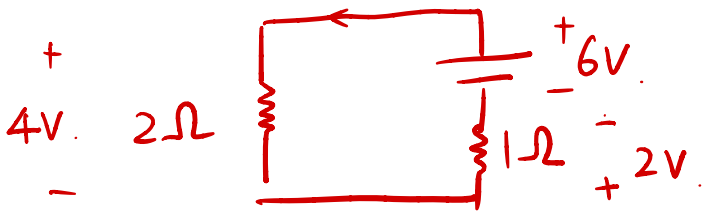
$d\vec{s}$ direction.

$$\psi_B = \iint_S \vec{B} \cdot d\vec{s} = \iint (1\hat{y}) \cdot (dx\,dz(-\hat{y})) = -\text{Area} = -2 \cdot 3t = -6t \ [\text{Wb}]$$

$$\mathcal{E} = -\frac{d\psi_B}{dt} = 6 \ [\text{V}]$$

Example: Motional EMF

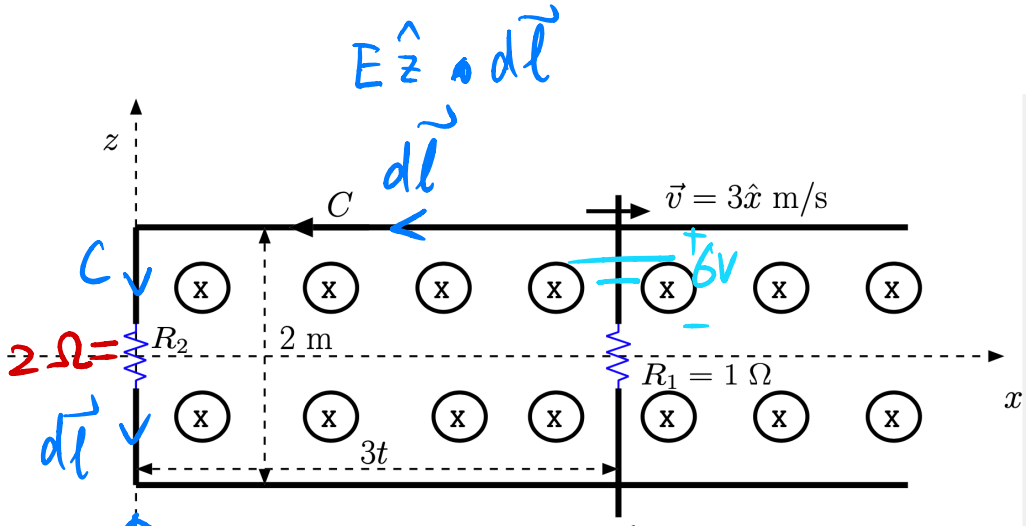
Given $R_2 = 2\Omega$, calculate electric field E_z.



$$\int_1^{-1} \vec{E} \cdot d\vec{\ell} = 4\,V$$

$$E_z \cdot (-2) = 4$$

$$E_z = -2\ V/m$$



$$6V = \int_1^{-1} \vec{E} \cdot d\vec{\ell} + \int_{-1}^{1} (\vec{E} + \vec{v} \times \vec{B}) \cdot d\vec{\ell}$$

$$\int_{-1}^{1} (\vec{E} + \vec{v} \times \vec{B}) \cdot d\vec{\ell} = 2V$$

$$3\hat{x} \times 1\hat{y}$$

$$2E_z + 6 = 2$$

$$E_z = -2\ V/m$$

Comparing Faraday and Ampere's Law

- Faraday's Law
  - Time varying magnetic fields generate emf (voltage)

$$\nabla\times\vec{\boldsymbol{E}} = -\frac{d\vec{\boldsymbol{B}}}{dt}$$

- Ampere's Law
  - Electric currents generate magnetic fields (Ampere's static law)

$$\nabla\times\vec{\boldsymbol{H}} = J$$

  - Time varying electric fields generate magnetic fields

$$\nabla\times\vec{\boldsymbol{H}} = J + \frac{d\vec{\boldsymbol{D}}}{dt}$$

EMF and MMF

$$\oint_C \vec{E} \bullet \vec{dl} = EMF$$

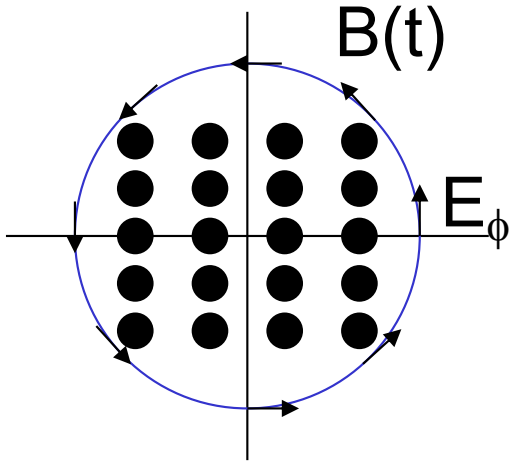
$$\oint_C \vec{H} \bullet \vec{dl} = MMF$$

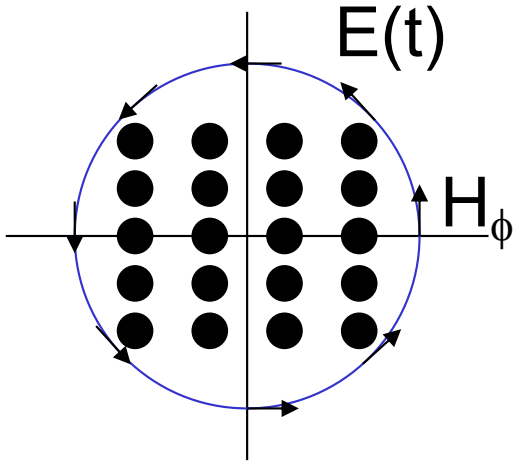
E=Volts/m

H=Amps/m

EMF=Volts

MMF=Amps





Lecture 14 Summary

Faraday's Law: $\nabla \times \vec{\boldsymbol{E}} = -\frac{d\vec{\boldsymbol{B}}}{dt}$

- The generated emf opposes the change of rate of B field and for a loop with N turns is N times larger
- The direction for C and dS determined by right hand rule

- Next class
  - Inductance